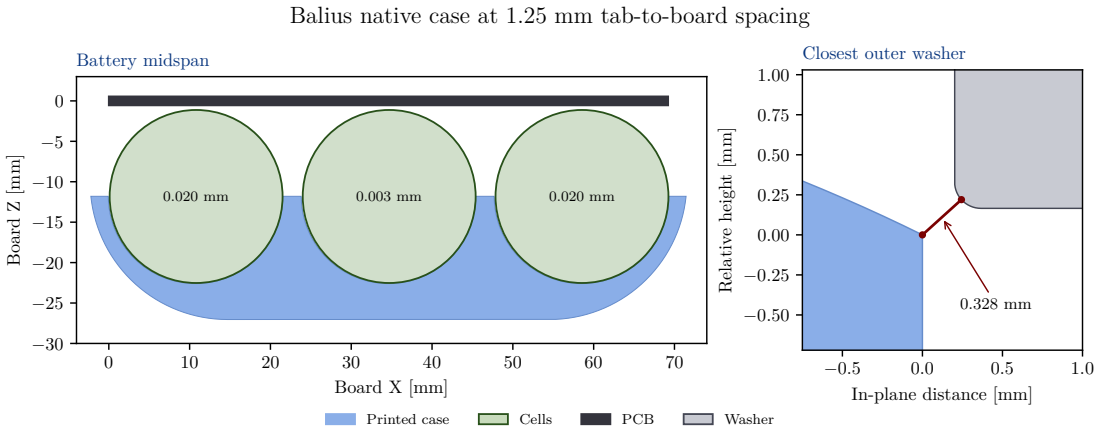
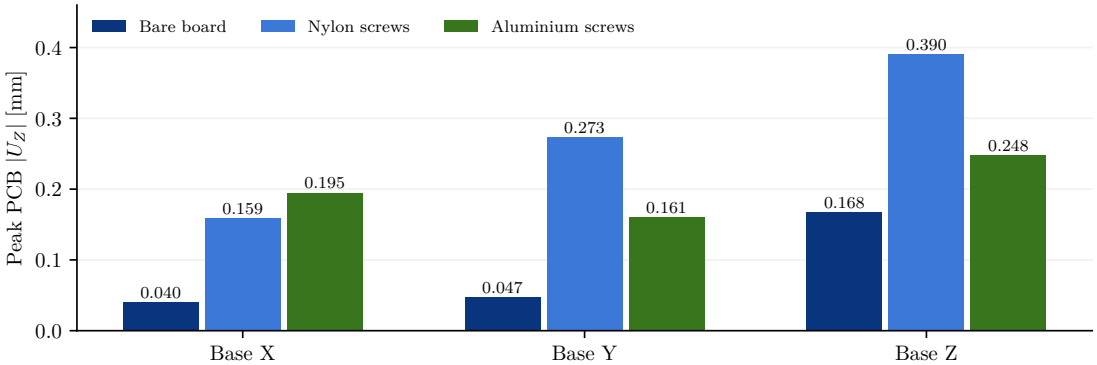
The three-pocket Balius case fits at 1.25 mm stand-off



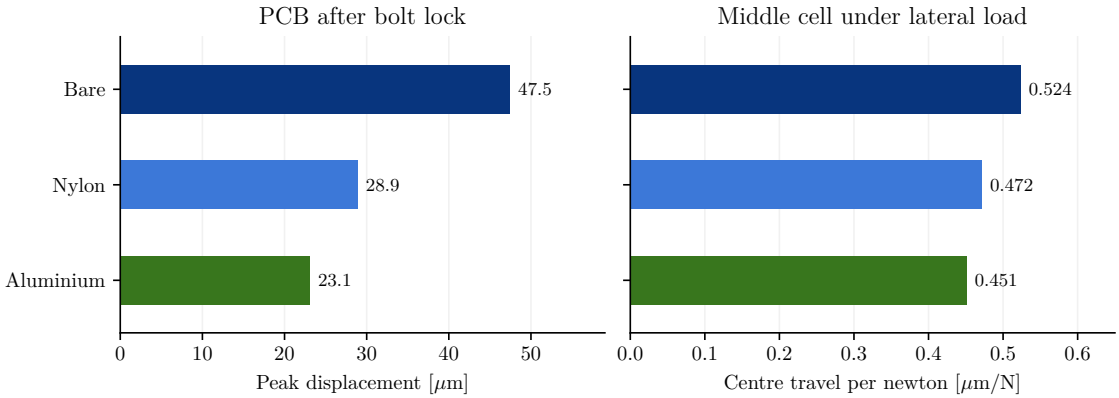
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

The snug case increases board travel in the seeded clips



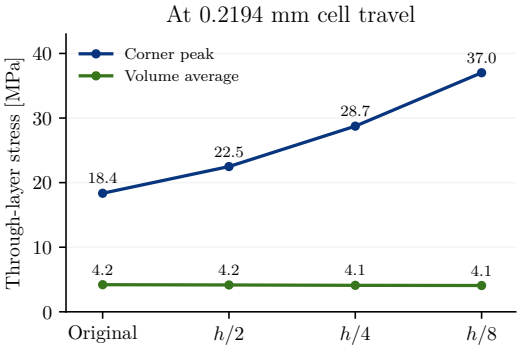
During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

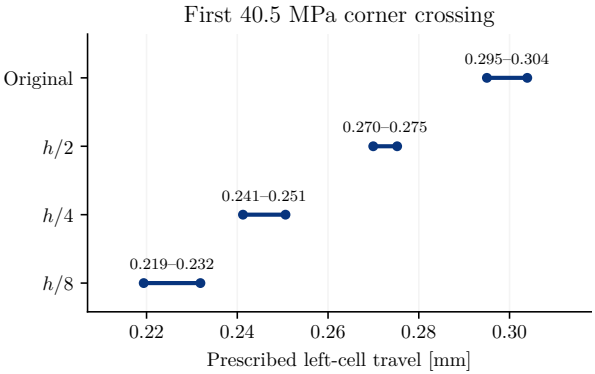
The case restrains the middle cell at bolt lock



Only the middle cell touched the case after tightening. In an independent calculation on the installed board, it moved **9.9% less with nylon screws** and **13.9% less with aluminium** under the same small lateral force. The finer nylon mesh gave 9.4%. At each outer cell the change from bare was below 0.36%. The PCB displacement at left comes from native ANSYS.

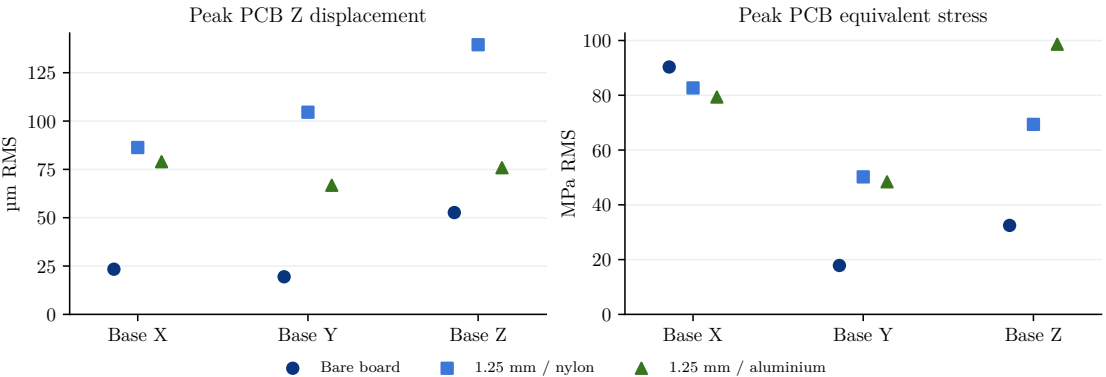
The tab-corner stress depends strongly on mesh size





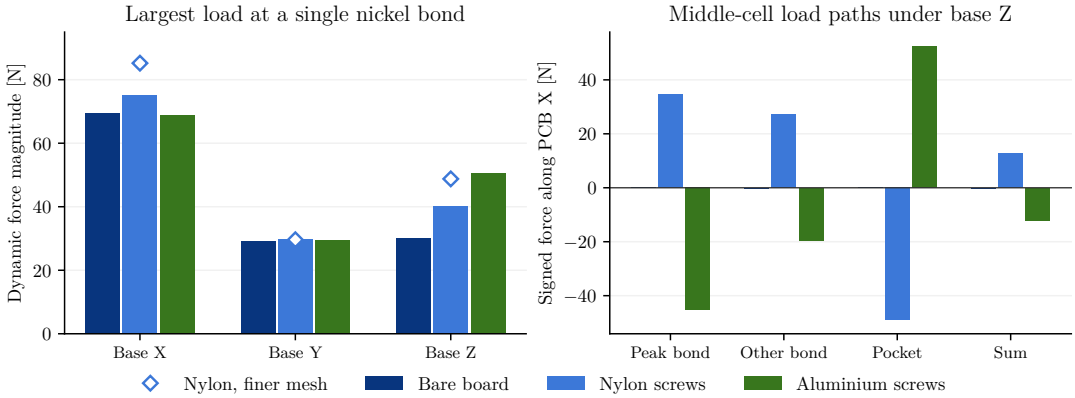
At **8.54 N**, cell travel fell from **0.500 mm bare** to **about 0.034 mm** with nylon screws. Under the same native boundary motions, local refinement moved the PLA corner crossing from about **0.30 to 0.23 mm**. The corner result has yet to converge.

Aluminium cuts board travel relative to nylon but raises base-Z stress



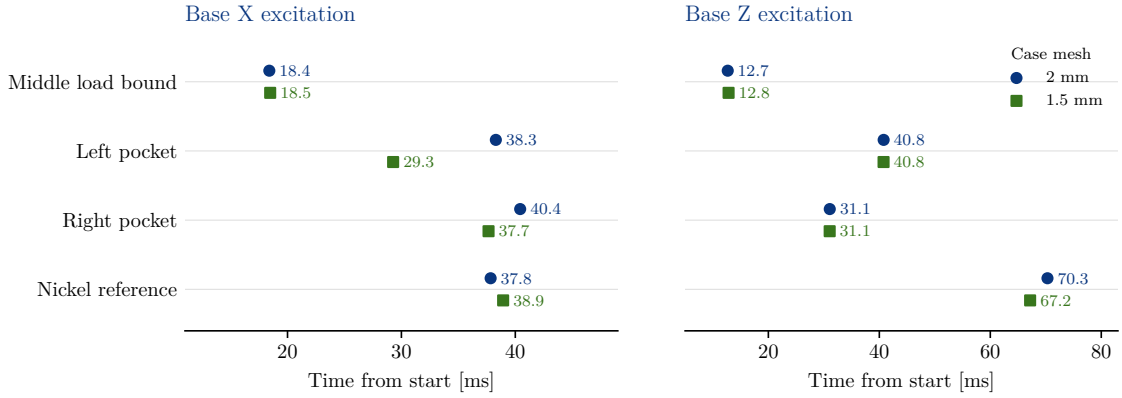
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

Board-normal vibration loads the middle cell's nickel bond



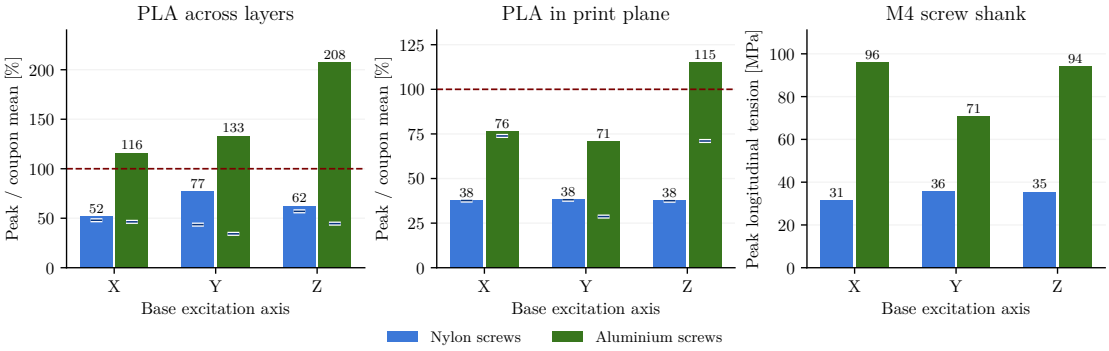
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

The case mesh changes which event comes first under base X



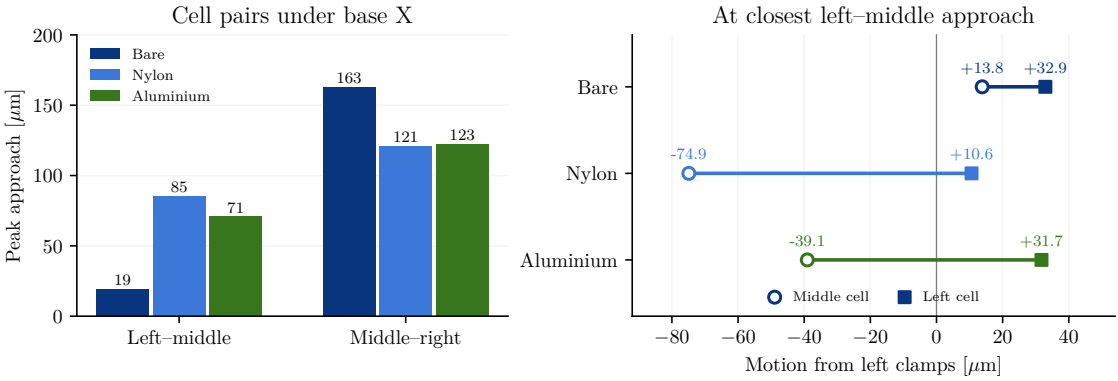
With nylon screws, refining the case moves left-pocket contact from **38.3 to 29.3 ms**. It crosses before the nickel reaches 185 MPa on the fine mesh. The nominal mesh gives the opposite order. Under base Z the right cell comes first on both nylon meshes. The nominal aluminium assembly reaches the left pocket first.

The printed tab exceeds its coupon reference with aluminium screws



On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.

Under base X, the left and middle cells move closer together



The left cell's peak travel relative to its clamps fell by **29% with nylon** and **12% with aluminium**. Rotation accounted for **82%** of that travel in all three assemblies. With the middle cell free to move, peak approach rose from **19 μm bare** to **85 and 71 μm**. Contact stays fixed at bolt lock in these native histories.